



Network Scanner Project Using Network Mapper

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Objective:

- To perform **Network reconnaissance and enumeration** using **Nmap**
- To identify:
 - Live hosts
 - Open ports and services
 - OS and device types
 - Vulnerable services
- To analyze findings from the perspective of:
 - Ethical Hacker
 - Red Team Analyst
 - VAPT Consultant
 - SOC Analyst
 - Incident Responder

Tools Used:

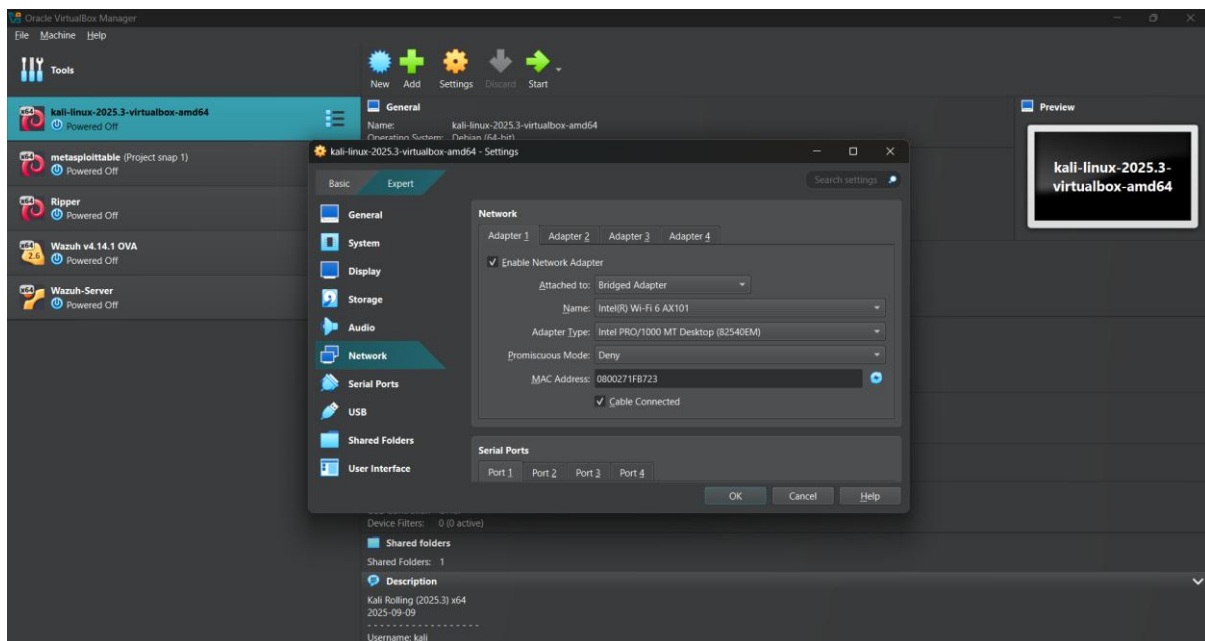
Tools	Purpose
Kali Linux	Penetration Testing OS
Nmap	Network scanning & enumeration
Oracle Virtual Box	Virtual lab environment
Target Machine (Metasploitable)	Authorized target
Snipping Tool	Evidence for report

Lab Setup & Network Architecture:

Lab Environment Setup:

The lab environment was configured using Oracle VirtualBox to simulate a real-world penetration testing environment. The attacker machine was Kali Linux, and the target machine was Metasploitable. Both systems were connected using Bridged Adapter networking to ensure isolated and authorized testing.

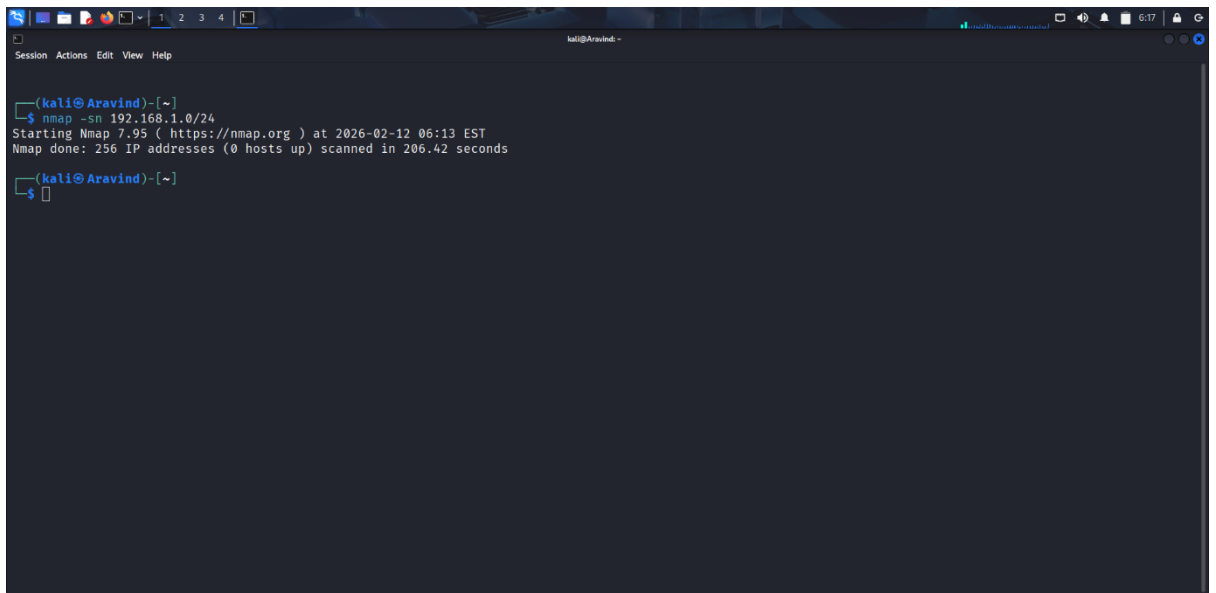
- Kali Linux IP address
- Metasploitable IP address
- Network type (NAT / Bridged / Host-only)
- VirtualBox configuration



Procedure:

1. Host Discovery (Live Systems):

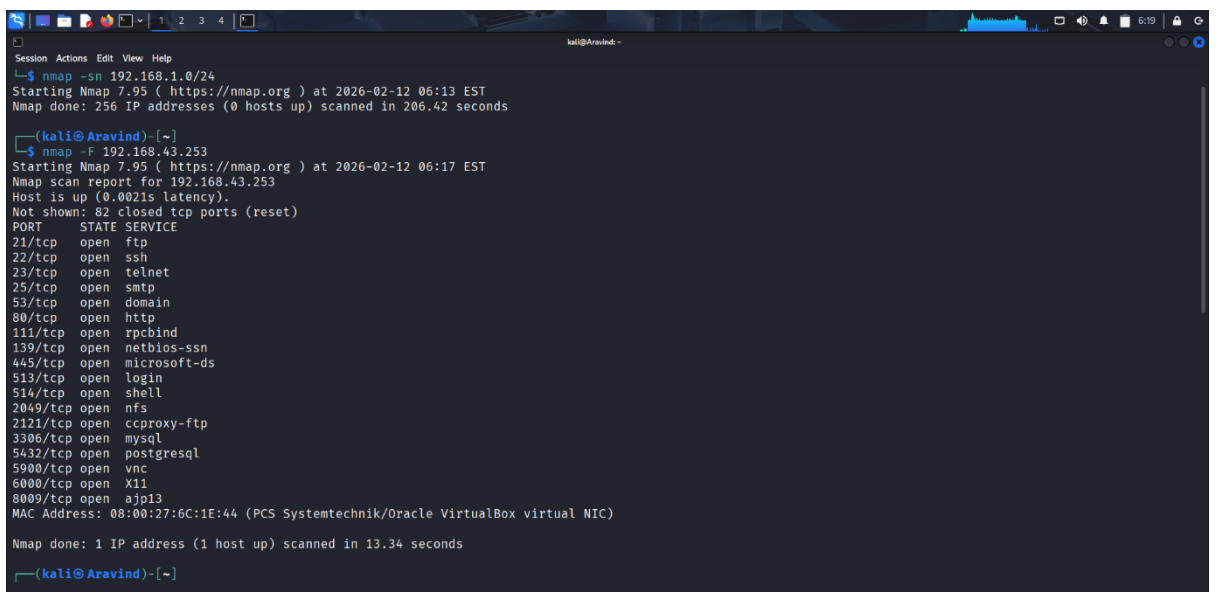
`nmap -sn 192.168.1.0/24` – Identifies live devices on the network



```
kali@Aravind: ~  
$ nmap -sn 192.168.1.0/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 06:13 EST  
Nmap done: 256 IP addresses (0 hosts up) scanned in 206.42 seconds  
$
```

2. Fast Port Scan:

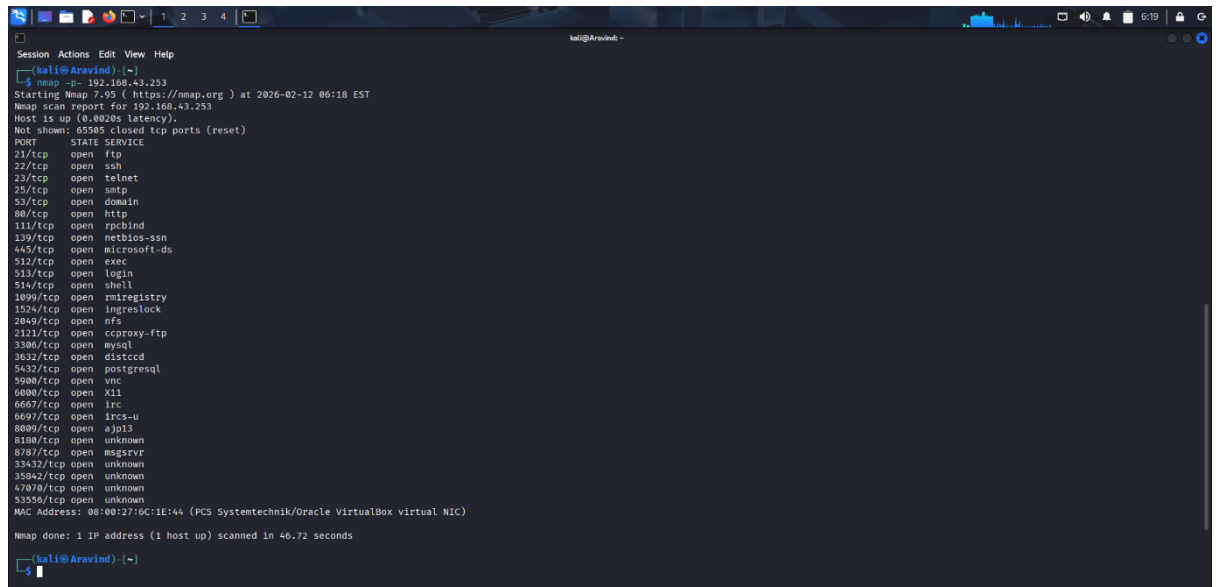
`nmap -F 192.168.43.253` – Quickly Identifies commonly exploited ports



```
kali@Aravind: ~  
$ nmap -sn 192.168.1.0/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 06:13 EST  
Nmap done: 256 IP addresses (0 hosts up) scanned in 206.42 seconds  
  
kali@Aravind: ~  
$ nmap -F 192.168.43.253  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 06:17 EST  
Nmap scan report for 192.168.43.253  
Host is up (0.0021s latency).  
Not shown: 82 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
513/tcp   open  login  
514/tcp   open  shell  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
8009/tcp  open  ajp13  
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 13.34 seconds  
$
```

3. Full TCP Port Scan:

`nmap -p- 192.168.43.253` – Detects hidden services attackers looks for



```
Session Actions Edit View Help
(kali@Aravind) ~
$ nmap -p- 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 06:18 EST
Nmap scan report for 192.168.43.253
Host is up (0.0020s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distcd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8188/tcp  open  unknown
8180/tcp  open  msgrvr
33432/tcp open  unknown
35842/tcp open  unknown
47070/tcp open  unknown
53558/tcp open  unknown
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 46.72 seconds
(kali@Aravind) ~
$
```

4. Service & Version Detection:

`nmap -sV 192.168.43.253` – Helps to exploit Matching Vulnerabilities



```
Session Actions Edit View Help
(kali@Aravind) ~
$ nmap -sV 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 06:19 EST
Nmap scan report for 192.168.43.253
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smb 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smb 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath gmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8009/tcp  open  ajp13        Apache Tomcat/Coyote JSP engine 1.1
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 67.25 seconds
(kali@Aravind) ~
$
```

5. OS Detection:

`nmap -O 192.168.43.253` – Determines Windows/Linux version for exploit selection



```
Session Actions Edit View Help
(kali@Aravind)-[~]
$ nmap -O 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 06:21 EST
Nmap scan report for 192.168.43.253
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccdproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.80 seconds
(kali@Aravind)-[~]
```

6. Aggressive Scan:

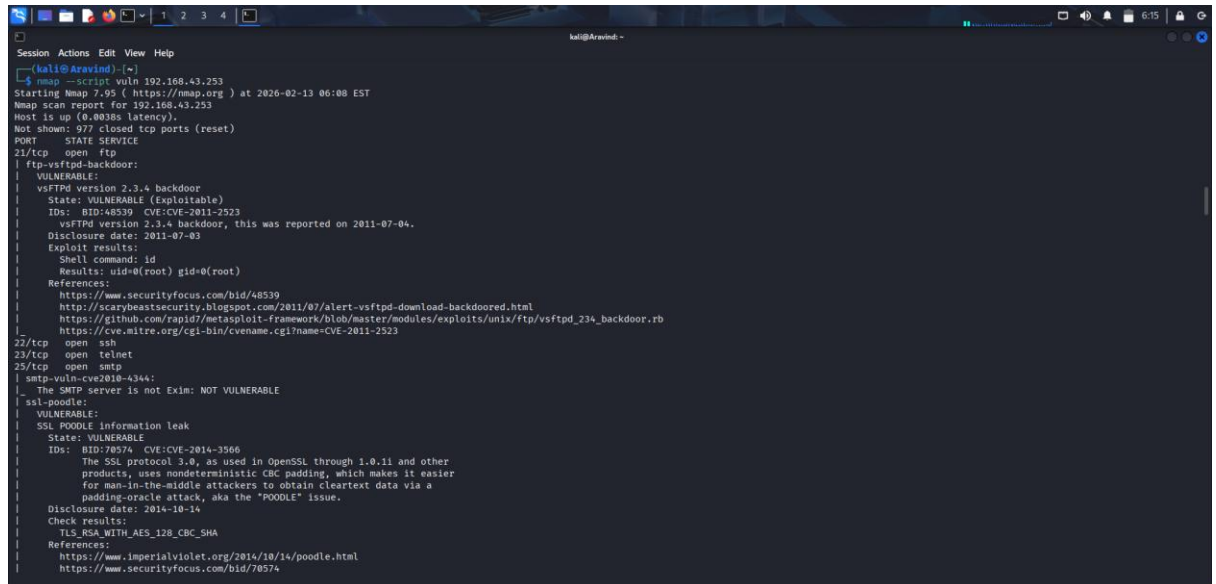
`nmap -A 192.168.43.253` – Includes OS, Version, Scripts, Traceroute



```
Session Actions Edit View Help
FTT min/avg/max/med = 2.681/3.241/4.731/0.862 ms
(kali@Aravind)-[~]
$ nmap -A 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:06 EST
Nmap scan report for 192.168.43.253
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  vsftpd 2.3.4
ftp-syst:
  STAI:
  FTP server status:
    Connected to 192.168.43.40
    Logged in as ftp
    TYPE: ASCII
    No session bandwidth limit
    Session timeout in seconds is 300
    Control connection is plain text
    Data connections will be plain text
    vsFTN 2.3.4 - secure, fast, stable
  End of status
  (ftp>anon) Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
  ssh-hostkey:
    1024 60:8f:c1:01:01:01:01:01:01:01:01:01:01:01:01:01 (DSA)
    2048 56:56:24:0f:21:1d:0e:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet
25/tcp    open  smtp
  postfix smtpd
  sslv2:
    sslv2 supported
  ciphers:
    SSL2_CCS_64_CBC_WITH_MD5
    SSL2_CCS_128_CBC_EXPORT40_WITH_MD5
    SSL2_CCS_128_CBC_EXPORT40_WITH_MD5
    SSL2_CCS_128_CBC_EXPORT40_WITH_MD5
    SSL2_CCS_128_CBC_EXPORT40_WITH_MD5
    SSL2_CCS_128_CBC_EXPORT40_WITH_MD5
    SSL2_CCS_128_CBC_EXPORT40_WITH_MD5
  ssl-date: 2026-02-13T11:07:17-04:00; vs from scanner time.
  ssl-cert: Subject: commonName=Aravind-Base.localdomain/organizationName=OCESA/stateOrProvinceName=There is no such thing outside US/countryName=XX
  Not valid before: 2018-03-17T14:07:14
  Not valid after: 2018-04-10T14:07:14
  cert-comments: metasploit@localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp    open  domain
  bind.version: 9.4.2
80/tcp    open  http
  http-title: MetasploitLab2 - Linux
  http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind
  rpcinfo:
    program version    port/proto service
    100000 2                    111/tcp  rpcbind
    100000 2                    111/udp  rpcbind
    100000 2.3.4                2049/tcp nfs
(kali@Aravind)-[~]
```

7. Vulnerability Scanning (NSE Scripts):

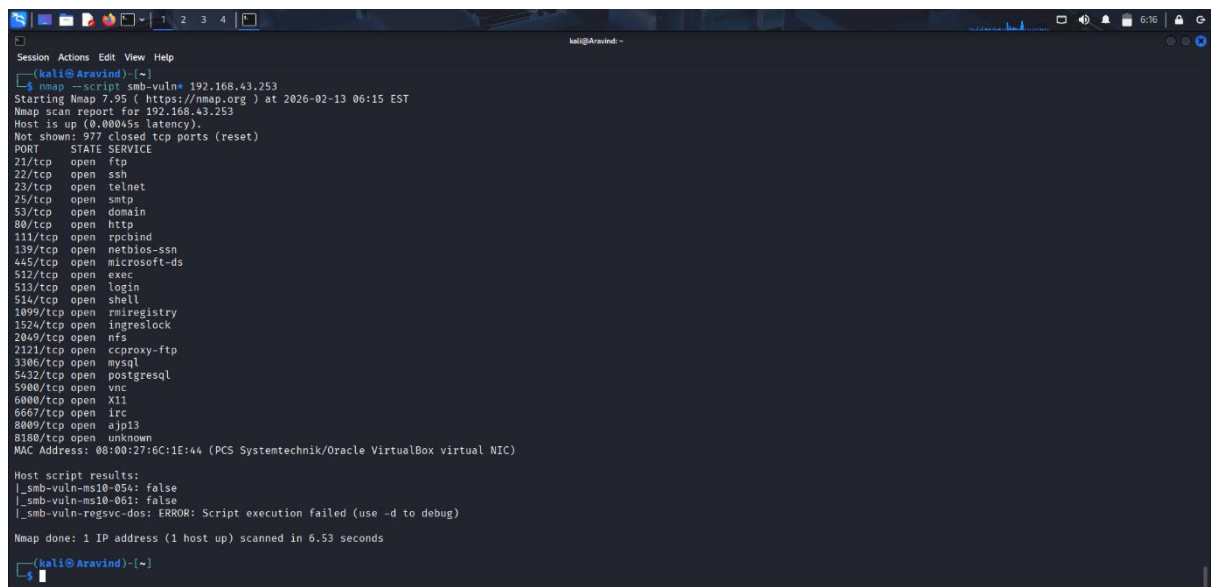
`nmap --script vuln 192.168.43.253` – Finds Known CVE's



```
Session Actions Edit View Help
(kali@Aravind)-[~]
$ nmap --script vuln 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:08 EST
Nmap scan report for 192.168.43.253
Host is up (0.0036s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
ftp-vsftpd-backdoor:
  VULNERABLE:
  vsftpd version 2.3.4 backdoor
  State: VULNERABLE (Exploitable)
  IDs: BID:48539 CVE:CVE-2011-2523
  vsftpd version 2.3.4 backdoor, this was reported on 2011-07-04.
  Disclosure date: 2011-07-03
  Exploit results:
  Shell command: id
  Results: uid=0(root) gid=0(root)
  References:
  https://www.securityfocus.com/bid/48539
  http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
  https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
  https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
smtp-vuln-cve2010-4344:
  The SMTP server is not Exim: NOT VULNERABLE
ssl-poodle:
  VULNERABLE:
  SSL POODLE Information leak
  State: VULNERABLE
  IDs: BID:70574 CVE:CVE-2014-3566
  The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
  products, uses nondeterministic CBC padding, which makes it easier
  for man-in-the-middle attackers to obtain cleartext data via a
  padding-oracle attack, aka the "POODLE" issue.
  Disclosure date: 2014-10-14
  Check results:
  TLS_RSA_WITH_AES_128_CBC_SHA
  References:
  https://www.imperialviolet.org/2014/10/14/poodle.html
  https://www.securityfocus.com/bid/70574
```

8. SMB Vulnerability Scan (Specific Vuln Scan):

`nmap --script smb-vuln 192.168.43.253` – Detects SMB exploits like EternalBlue



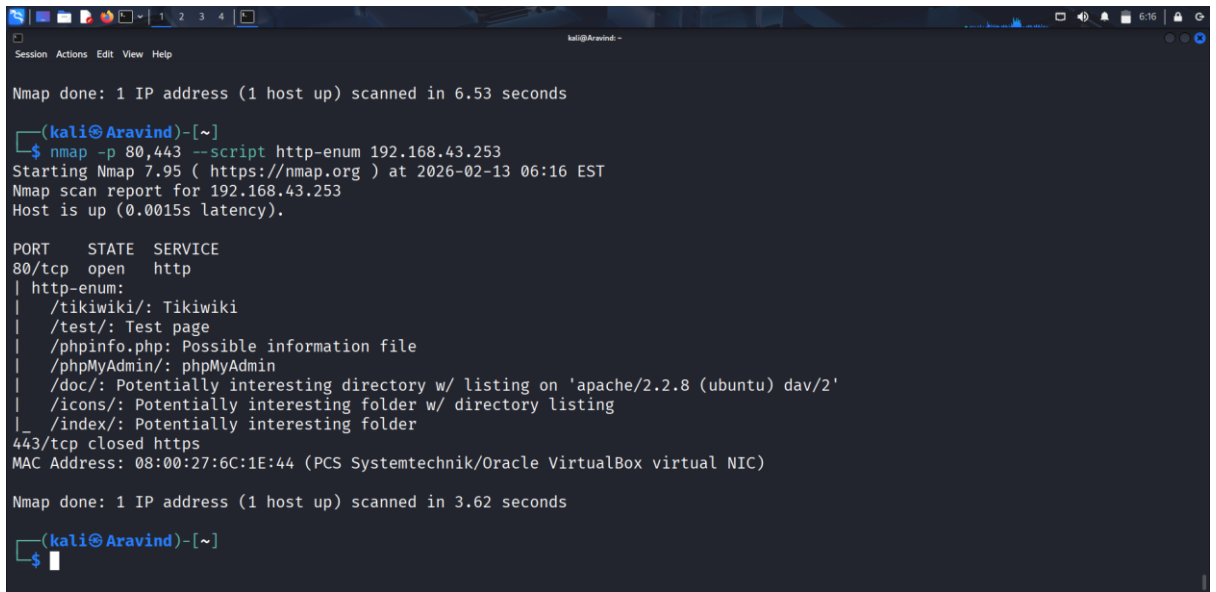
```
Session Actions Edit View Help
(kali@Aravind)-[~]
$ nmap --script smb-vuln 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:15 EST
Nmap scan report for 192.168.43.253
Host is up (0.00045s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: false
|_smb-vuln-rgsvnc-dos: ERROR: Script execution failed (use -d to debug)

Nmap done: 1 IP address (1 host up) scanned in 6.53 seconds
(kali@Aravind)-[~]
$
```

9. Web Service Enumeration:

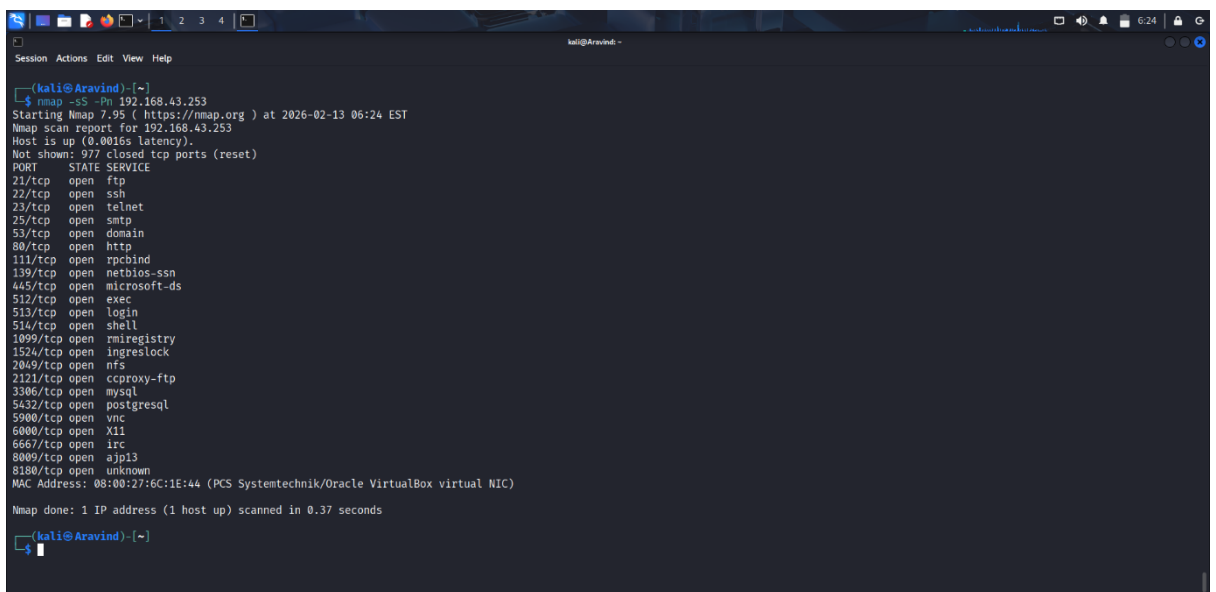
`nmap -p 80,443 -script http-enum 192.168.43.253` – Website Reconnaissance



```
kali@Aravind: ~  
Nmap done: 1 IP address (1 host up) scanned in 6.53 seconds  
  
(kali@Aravind)-[~]  
$ nmap -p 80,443 --script http-enum 192.168.43.253  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:16 EST  
Nmap scan report for 192.168.43.253  
Host is up (0.0015s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
| http-enum:  
| /tikiwiki/: Tikiwiki  
| /test/: Test page  
| /phpinfo.php: Possible information file  
| /phpMyAdmin/: phpMyAdmin  
| /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'  
| /icons/: Potentially interesting folder w/ directory listing  
|_ /index/: Potentially interesting folder  
443/tcp    closed https  
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 3.62 seconds  
  
(kali@Aravind)-[~]  
$
```

10. Firewall & IDS Evasion:

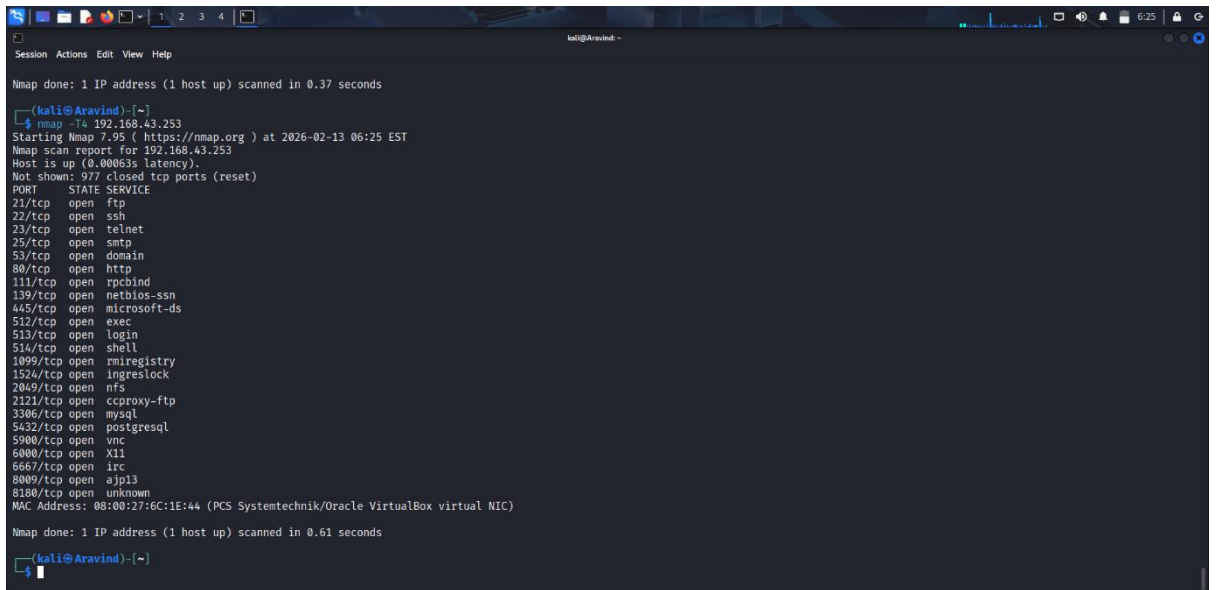
`nmap -sS -Pn 192.168.43.253` – Simulates attacker stealth techniques



```
kali@Aravind: ~  
$ nmap -sS -Pn 192.168.43.253  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:24 EST  
Nmap scan report for 192.168.43.253  
Host is up (0.0016s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp    open  rpcbind  
139/tcp    open  netbios-ssn  
445/tcp    open  microsoft-ds  
512/tcp    open  exec  
513/tcp    open  login  
514/tcp    open  shell  
1099/tcp   open  rmiregistry  
1524/tcp   open  ingreslock  
2049/tcp   open  nfs  
2121/tcp   open  ccproxy-ftp  
3306/tcp   open  mysql  
5432/tcp   open  postgresql  
5900/tcp   open  vnc  
6000/tcp   open  X11  
6667/tcp   open  irc  
8009/tcp   open  ajp13  
8180/tcp   open  unknown  
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.37 seconds  
  
(kali@Aravind)-[~]  
$
```


11. Scan Timing Analysis:

`nmap -T4 192.168.43.253` – Helps to detect aggressive scanning in a particular time



```
Nmap done: 1 IP address (1 host up) scanned in 0.37 seconds

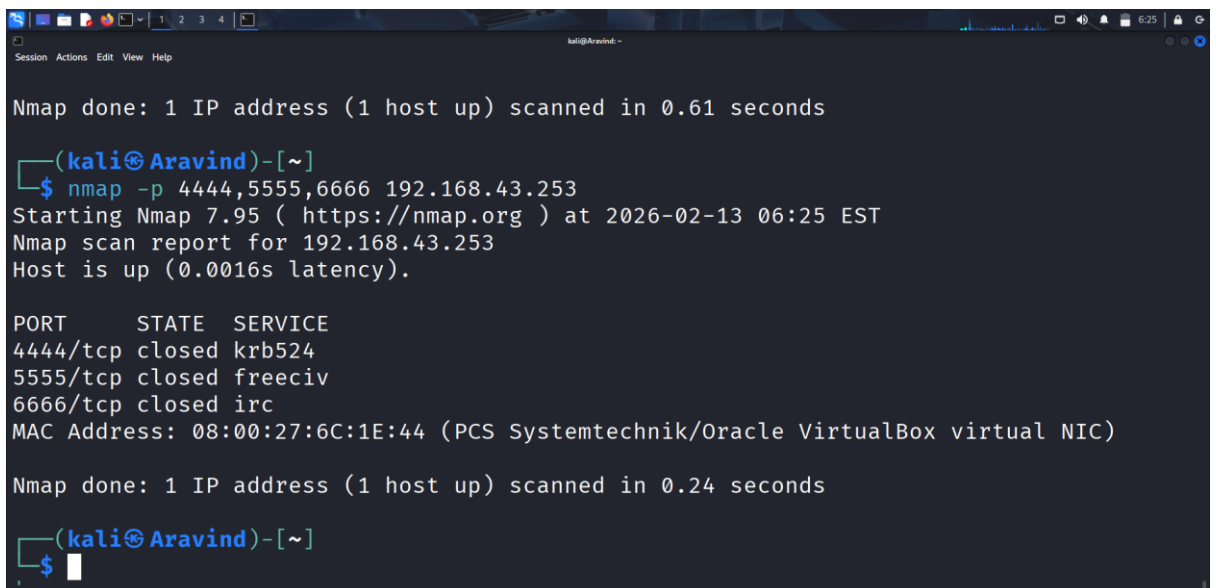
(kali@Aravind)-[~]
$ nmap -T4 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:25 EST
Nmap scan report for 192.168.43.253
Host is up (0.00063s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ceph-proxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.61 seconds

(kali@Aravind)-[~]
$
```

12. Detects Backdoor Ports:

`nmap -p 4444,5555,6666 192.168.43.253` – Common malware C2 ports



```
Nmap done: 1 IP address (1 host up) scanned in 0.61 seconds

(kali@Aravind)-[~]
$ nmap -p 4444,5555,6666 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:25 EST
Nmap scan report for 192.168.43.253
Host is up (0.0016s latency).

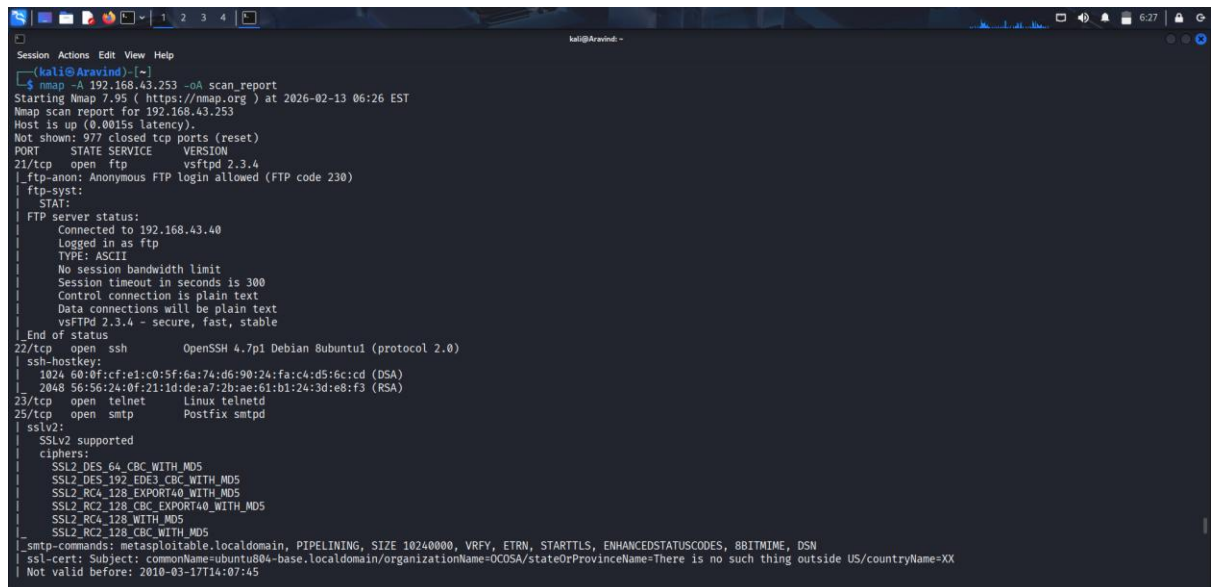
PORT      STATE SERVICE
4444/tcp   closed krb524
5555/tcp   closed freeciv
6666/tcp   closed irc
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

(kali@Aravind)-[~]
$
```

13. Output for Logs & Evidence:

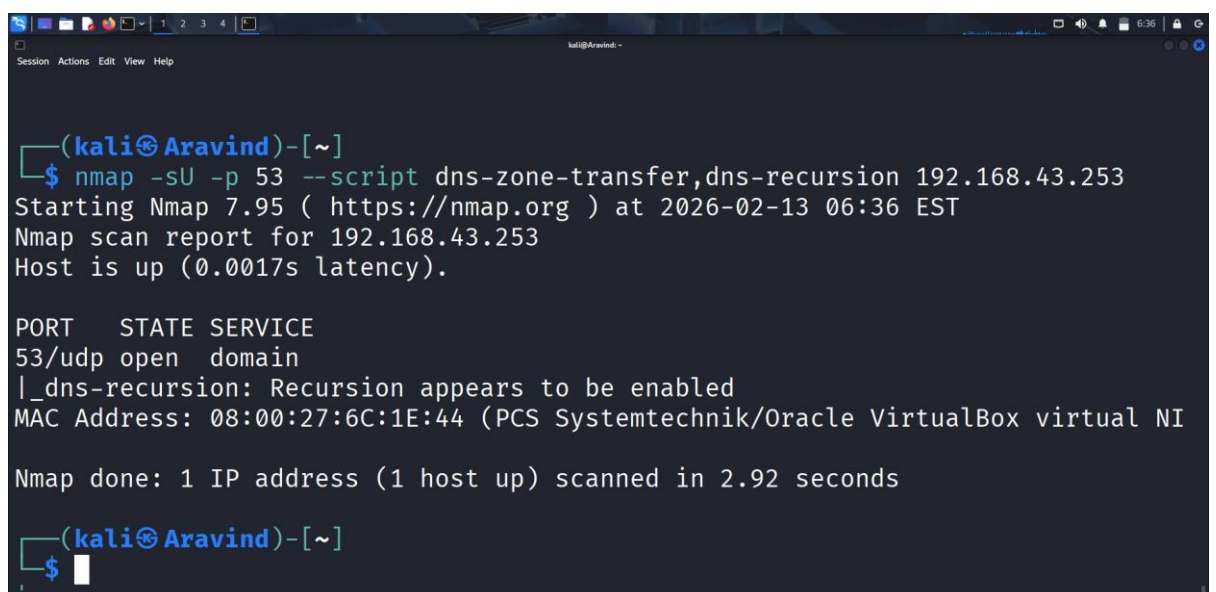
`nmap -A 192.168.43.253 -oA scan_report` – Generates: .nmap, .xml, .gnmap reports used in investigations



```
(kali@Aravind)-[~]
$ nmap -A 192.168.43.253 -oA scan_report
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:26 EST
Nmap scan report for 192.168.43.253
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.43.40
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_  SSL2_DES_64_CBC_WITH_MD5
|_  SSL2_DES_192_EDE3_CBC_WITH_MD5
|_  SSL2_RC4_128_EXPORT40_WITH_MD5
|_  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_  SSL2_RC4_128_WITH_MD5
|_  SSL2_RC2_128_CBC_WITH_MD5
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
|_ssl-cert: Subject: commonName=rubuntu04-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
```

14. DNS Enumeration:

`nmap -sU -p 53 - --script dns-zone-transfer ,dns-recursion 192.168.43.253`
– Can able to Reveal Internal domain records, subdomains, mail servers.



```
(kali@Aravind)-[~]
$ nmap -sU -p 53 --script dns-zone-transfer,dns-recursion 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:36 EST
Nmap scan report for 192.168.43.253
Host is up (0.0017s latency).

PORT      STATE SERVICE
53/udp    open  domain
|_dns-recursion: Recursion appears to be enabled
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NI)

Nmap done: 1 IP address (1 host up) scanned in 2.92 seconds

(kali@Aravind)-[~]
$
```

15. Firewall Detection + Packet Fragmentation:

`nmap -sS -Pn -f - -reason 192.168.43.253` – Skips host discovery and bypass basic firewall shows why port state was determined



```
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 0.43 seconds

(kali@Aravind)-[~]
$ nmap -sS -Pn -f - -reason 192.168.43.253
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-13 06:42 EST
Nmap scan report for 192.168.43.253
Host is up, received arp-response (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      REASON
21/tcp    open  ftp          syn-ack ttl 64
22/tcp    open  ssh          syn-ack ttl 64
23/tcp    open  telnet       syn-ack ttl 64
25/tcp    open  smtp         syn-ack ttl 64
53/tcp    open  domain       syn-ack ttl 64
80/tcp    open  http         syn-ack ttl 64
111/tcp   open  rpcbind      syn-ack ttl 64
139/tcp   open  netbios-ssn  syn-ack ttl 64
445/tcp   open  microsoft-ds syn-ack ttl 64
512/tcp   open  exec         syn-ack ttl 64
513/tcp   open  login        syn-ack ttl 64
514/tcp   open  shell        syn-ack ttl 64
1099/tcp  open  rmiregistry  syn-ack ttl 64
1524/tcp  open  ingreslock   syn-ack ttl 64
2049/tcp  open  nfs          syn-ack ttl 64
2121/tcp  open  cproxy-ftp   syn-ack ttl 64
3306/tcp  open  mysql        syn-ack ttl 64
5432/tcp  open  postgresql   syn-ack ttl 64
5900/tcp  open  vnc          syn-ack ttl 64
6000/tcp  open  X11          syn-ack ttl 64
6667/tcp  open  irc          syn-ack ttl 64
8009/tcp  open  ajp13        syn-ack ttl 64
8180/tcp  open  unknown     syn-ack ttl 64
MAC Address: 08:00:27:6C:1E:44 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.17 seconds

(kali@Aravind)-[~]
$
```

Result Analysis:

1. Host Discovery Analysis (nmap -sn)

The host discovery scan confirmed the availability of active systems within the subnet. The Metasploitable machine responded to ICMP requests, indicating that it is discoverable without restriction.

Enterprise Risk:

In real corporate environments, unrestricted ICMP responses allow attackers to map internal networks easily during reconnaissance.

Recommendation:

- Implement network segmentation.
- Restrict ICMP responses at firewall level.
- Monitor abnormal ping sweep behavior via SIEM.

2. Port Exposure Analysis (-F and -p-)

The fast scan identified commonly exposed services, while the full TCP scan revealed multiple open ports, including legacy and insecure services.

Observed Enterprise Risks:

- Multiple open ports increase the attack surface.
- Services running on non-standard ports may indicate misconfiguration or hidden services.
- Legacy services such as Telnet and FTP are insecure and transmit credentials in plaintext.

Business Impact:

An attacker gaining access through one exposed service can escalate privileges and move laterally within the organization.

Recommendation:

- Close unused ports.
- Enforce least privilege exposure.
- Replace insecure protocols with secure alternatives (e.g., SSH instead of Telnet).

3. Service & Version Enumeration (-sV)

Service version detection provided detailed software fingerprints.

Enterprise Significance:

Version disclosure enables attackers to:

- Map services to known CVEs.
- Identify outdated or unsupported software.
- Plan targeted exploitation.

Risk Level:

High — If patch management is weak, known exploits can be weaponized rapidly.

Recommendation:

- Implement automated patch management.
 - Disable version banners where possible.
 - Conduct periodic vulnerability scans.
-

4. OS Fingerprinting (-O)

Operating system detection successfully identified the Linux distribution used by Metasploitable.

Enterprise Risk:

OS fingerprinting simplifies exploit selection and payload compatibility for attackers.

Defensive Insight:

Modern enterprise networks deploy IDS/IPS systems to detect OS fingerprinting attempts.

5. Aggressive Scan Findings (-A)

The aggressive scan provided a consolidated view of:

- OS details
- Service versions
- Default script outputs
- Network path information

Enterprise Observation:

This demonstrates how a single automated command can reveal a full attack surface within minutes.

Risk:

If perimeter defenses fail to detect aggressive scans, attackers can perform complete reconnaissance without triggering alerts.

6. Vulnerability Script Analysis (--script vuln)

The vulnerability scripts identified services potentially associated with publicly disclosed vulnerabilities.

Enterprise Impact:

- Known CVEs can lead to Remote Code Execution.
- Systems without patch governance are high-risk.
- Exploitable services may allow full system compromise.

Risk Classification:

Critical — If externally accessible.

7. SMB Vulnerability Analysis (smb-vuln)

SMB enumeration identified weaknesses in file-sharing services.

Enterprise Risk:

SMB vulnerabilities are frequently used in:

- Ransomware propagation
- Lateral movement attacks
- Credential harvesting

Real-World Example:

Worm-based attacks leverage SMB misconfigurations to spread across networks rapidly.

Recommendation:

- Disable SMBv1.
 - Restrict SMB access via firewall rules.
 - Apply Microsoft security patches promptly.
-

8. Web Service Enumeration (http-enum)

Web enumeration revealed accessible directories and exposed web services.

Enterprise Risk:

Exposed web applications are common entry points for:

- SQL Injection
- Remote File Inclusion
- Cross-Site Scripting

Mitigation:

- Conduct regular web application penetration testing.
 - Remove unnecessary directories.
 - Implement WAF protection.
-

9. Firewall or Stealth & Evasion Testing (-sS -Pn)

The SYN stealth scan bypassed basic host discovery techniques.

Enterprise Insight:

If undetected, this indicates:

- Weak IDS/IPS configuration
- Insufficient traffic monitoring

SOC teams must monitor SYN scan patterns and unusual connection attempts.

10. Scan Timing Analysis (-T4)

Aggressive timing demonstrates how quickly reconnaissance can be executed.

SOC Perspective:

Rapid scan patterns should trigger automated alerts in SIEM systems.

Failure to detect such activity increases dwell time for attackers.

11. Backdoor Port Scan (4444, 5555, 6666)

Scanning commonly used reverse-shell ports tests for compromise indicators.

Incident Response Perspective:

Unexpected open high-numbered ports may indicate:

- Malware infection
- Unauthorized services
- Command & Control channels

Regular network audits are critical.

12. DNS Enumeration (dns-zone-transfer, dns-recursion)

DNS testing attempted to extract internal records.

Enterprise Risk:

Successful zone transfers expose:

- Internal infrastructure mapping
- Mail servers
- Subdomains
- Internal IP ranges

This significantly accelerates attacker reconnaissance.

13. Firewall Evasion & Fragmentation (-f --reason)

Packet fragmentation attempts to bypass basic firewall filtering.

Enterprise Security Insight:

If fragmented packets are accepted without inspection, perimeter defenses may be misconfigured.

Modern NGFWs should reassemble fragmented packets for deep inspection.

Conclusion:

This network reconnaissance assessment conducted against the intentionally vulnerable **Metasploitable** system demonstrates how exposed services, outdated software versions, and weak configurations significantly increase an organization's attack surface.

The series of Nmap scans simulated real-world attacker reconnaissance techniques, including stealth scanning, vulnerability mapping, service fingerprinting, and firewall evasion testing. The results highlight that:

- Multiple exposed ports expand entry points.
- Legacy and unpatched services create critical exploitation risks.
- Inadequate firewall rules allow detailed enumeration.
- DNS and SMB misconfigurations can enable large-scale compromise.
- Rapid scan techniques may go undetected without proper monitoring.

From an enterprise cybersecurity perspective, the findings emphasize the importance of:

- Continuous vulnerability management
- Network segmentation
- Patch governance programs
- Deployment of IDS/IPS and SIEM monitoring
- Service hardening and secure configuration baselines

Overall, this project demonstrates how Nmap can replicate real adversarial reconnaissance techniques used in red teaming and penetration testing. It reinforces the necessity for proactive security controls and continuous monitoring to prevent exploitation and minimize organizational risk.